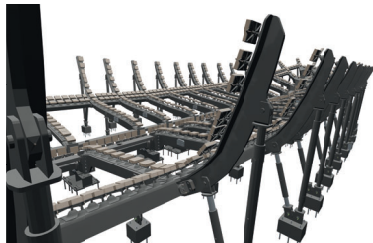


Navigating technical requirements and ethical dilemmas in a large-scale, ongoing interdisciplinary preservation project

Collections Unit, The Vasa Museum, Swedish National Maritime Transport and Military Museums, Stockholm, Sweden

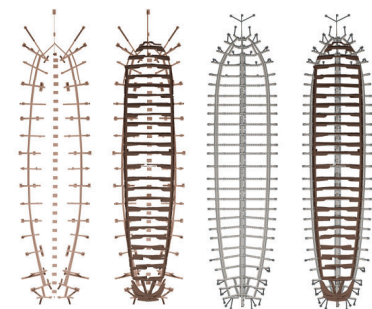
malin.sahlstedt@statensmuseermtf.se, hakan.thoren@statensmuseermtf.se

The 17th-century warship *Vasa* is currently undergoing the largest preservation effort since its raising and conservation in the 1960s and 1970s. The design and implementation of a new, improved ship support system were triggered by visible deformation of the ship's structure. The project became possible only after solid research and the dedication of an interdisciplinary group working closely with museum leadership.



Detail of the new outer support. Illustration Vasamuseet/SMMTE. CC-BY

The new system comprises 27 adjustable cradles connected via supports under the keel and linked through the hull to an internal onboard support structure. The design of both the external and internal support structures was developed in a 3D environment using laser scanning and photogrammetry of the ship as the starting point. Unlike the original support cradles, which were positioned based on the load-bearing capacity of the museum building's lower level, the new cradles are placed according to the ship's structural needs and load distribution. Reinforcements in the building have been made where necessary.



Old support Old support
ship's rider

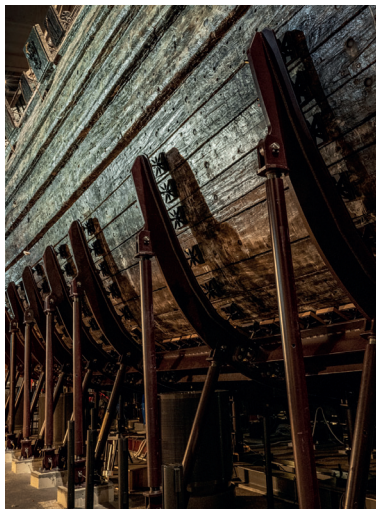


The new outer and inner support. Illustration Vasamuseet/SMMTF. CC-BY



Assembly of the new support structure. Photo Anneli Karlsson, Vasamuseet/SMMTE. CC-BY

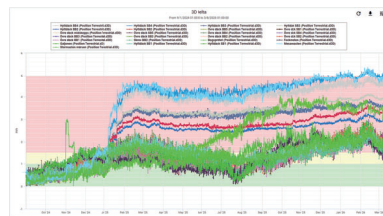
The adjustability of all major components will allow the ship – currently tilting about 1.5° to port – to be gradually shifted into a more upright, stable position.



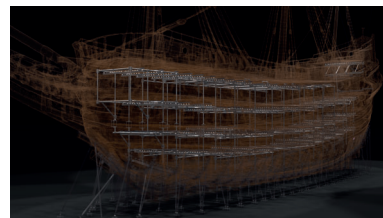
The port side of Vasa with the new support cradles in place. Photo Anneli Karlsson, Vasamuseet/SMMTE. CC-BY

The project, jointly led by the museum and a consulting firm, is highly dependent on specialised engineering and construction expertise. Conflicts of interest have been numerous, as technical requirements have collided with acceptable changes in the ship as an archaeological source and cultural historical object, and material choices with museum aesthetics and visitor experience. Ethical debates have addressed questions such as the number of holes that can justifiably be drilled through the hull and whether to cut through or remove planking to allow the passage of supports through the ship's decks. Full-scale on-site testing prior to new assembly phases has been a critical decision-making tool.

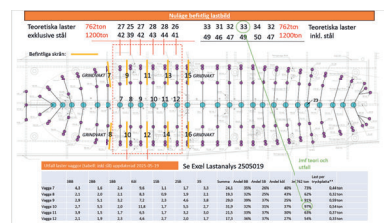
A laser tracker ensures the high-level precision required during support assembly and enables fast, accurate local deformation monitoring. In parallel, a total station survey analyses 30 points across the ship's upper structure every 15 minutes to provide global movement data. Each cradle is fitted with load cells to monitor how weight is distributed across the ship. This combination of local, global and load data is being used to guide the assembly process in real time. To date, changes in the position of the keel and hull remain within a 2 mm range. Some slightly larger deformations have been noted in the upper stern area but can be corrected in later phases of the assembly.



The global monitoring system indicates overall structural stability, as the displacements remain within a 2 mm threshold. Illustration Vasamuseet/SMMTF. CC-BY



Future onboard support structure. Illustration Vasemuseet/SMMTE. CC-BY



Full-scale testing results from the first part of the assembly of the outer support. Data from load cells on the new cradles show the ship's actual weight distribution, previously only theretically calculated.

The work of an interdisciplinary team of engineers, conservators, shipwrights, architects and curators, in an iterative process of learning, open dialogue and a willingness to revisit earlier decisions as theory met practice, has guided the assembly of the external support to its halfway point, so far on time and within the budget. Collaboration, compromise and collective dedication have proven key success factors in navigating the complex preservation challenges at the Vasa Museum.

Selected publications on the strength and mechanical behaviour of Vasa wood and ship structures:

Björkqvist, G., Norbäck, S., Jönsson, I. and Varma, K. 2016, 'Predicting the tensile strength of from IT-R contaminated archaeological wood from the 17th century Vasa ship: a study of degradation in recent oak and larca oak', *Leifnargangur* 2016, 855-65.

Björkqvist, I., Halonen, H., Lidfors, L.-E., Järven, T., Gansselt, E.-K. and Björklund, L. 2020, 'Status of degradation in archaeological oak from the 17th century Vasa ship: substantial strength loss correlates with reduction in (holocellulose molecular weight)', *Biomacromolecules* 21, 1521-7.

Brace, J., Alhar, M., Jönsson, A., Bommer, F. and Gansselt, E.-K. 2014, 'Towards determination of local and overall displacements of the Vasa ship structure: effect of its mechanical connections', in *The Royal Institution of Naval Architects Historic Ships Conference*, London: The Royal Institution of Naval Architects.

Langkilde, J. 2006, *Structure and Properties of Vasa Oak*, Licentiate thesis, Department of Aeronautical and Vehicle Engineering, Royal Institute of Technology, Stockholm.

Vorobyev, A. 2012, *Status and Time-Dependent Mechanical Behaviour of Preserved Archaeological Wood*, Case Studies of the Seventeenth Century Vasa Ship, PhD dissertation, Department of engineering Sciences, Applied Mechanics, Faculty of Science and Technology, Uppsala University.

Ward, L., Armand, G., Gansselt, E.-K., Björkqvist, I., Radzian, L. and Gansselt, E.-K. 2016, 'The influence of chemical degradation and polyethylene glycol on moisture-dependent cell wall properties of archaeological wood components: a case study of the Vasa ship', *Wood Research* 31, 1149-23.

